

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 4

This homework is due Friday, September 25, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Euler’s Formula and Trigonometric Identities

The famous Euler’s Formula is defined as follows:

$$e^{it} = \cos(t) + i \sin(t)$$

(a) Prove Euler’s Formula by starting from the Taylor expansion of e^θ .

Hint: recall the following Taylor expansions:

$$e^\theta = \sum_{n=0}^{\infty} \frac{\theta^n}{n!} = 1 + \theta + \frac{\theta^2}{2!} + \frac{\theta^3}{3!} + \dots$$

$$\cos(t) = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots$$

$$\sin(t) = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots$$

You will never have to derive these in this class, but hopefully they look familiar.

Solution: Substituting $\theta = it$ into the Taylor expansion of e^θ gives

$$\begin{aligned} e^{it} &= 1 + it + \frac{(it)^2}{2!} + \frac{(it)^3}{3!} + \frac{(it)^4}{4!} + \frac{(it)^5}{5!} + \dots \\ &= 1 + it - \frac{t^2}{2!} - \frac{it^3}{3!} + \frac{t^4}{4!} + \frac{it^5}{5!} + \dots \\ &= \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots\right) + i \left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots\right) \\ &= \cos(t) + i \sin(t). \end{aligned}$$

(b) Use Euler’s Formula to prove the Double Angle Identity:

$$\cos(2t) = \cos^2(t) - \sin^2(t)$$

Solution: First notice that $e^{i(2t)} = (e^{it})^2$. Expanding the right-hand side,

$$\begin{aligned}(e^{it})^2 &= (\cos(t) + i \sin(t))^2 \\ &= \cos^2 t + 2i \sin(t) \cos(t) + i^2 \sin^2 t \\ &= (\cos^2 t - \sin^2 t) + i(2 \sin(t) \cos(t)).\end{aligned}$$

Euler's formula also gives

$$e^{i(2t)} = \cos(2t) + i \sin(2t).$$

Equating real parts yields

$$\cos(2t) = \cos^2 t - \sin^2 t.$$

(c) Use Euler's Formula to prove the Angle Sum Identity:

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)$$

Solution: Using Euler's formula,

$$\begin{aligned}e^{i(\alpha+\beta)} &= e^{i\alpha} e^{i\beta} \\ &= (\cos(\alpha) + i \sin(\alpha)) (\cos(\beta) + i \sin(\beta)) \\ &= \cos(\alpha) \cos(\beta) + i \cos(\alpha) \sin(\beta) + i \sin(\alpha) \cos(\beta) + i^2 \sin(\alpha) \sin(\beta) \\ &= (\cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)) + i(\sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta)).\end{aligned}$$

Euler's formula also gives

$$e^{i(\alpha+\beta)} = \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

Equating real parts yields

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta).$$

(d) Use Euler's Formula to prove the hyperbolic cosine function identity, denoted $\cosh(t)$:

$$\cosh(t) = \frac{e^t + e^{-t}}{2}.$$

This pops up all the time in physics and engineering! For example, the shape of a chain hanging under its own weight is modeled with $\cosh$. Justify the name of this function.

Hint: recall inverse Euler's identity, and make a judicious (perhaps imaginary) choice of argument.

Solution: Inverse Euler's identity gives

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}.$$

Evaluating at the imaginary argument it gives

$$\begin{aligned}\cos(it) &= \frac{e^{i(it)} + e^{-i(it)}}{2} \\ &= \frac{e^{-t} + e^t}{2} \\ &= \cosh(t).\end{aligned}$$

Thus, hyperbolic cosine is ordinary cosine evaluated at an imaginary argument.

2. Roots of Unity and the Fourier Basis

Consider throughout this problem a fixed positive integer p . We define the fundamental frequency

$$\omega_0 = \frac{2\pi}{p},$$

and the p -th root of unity as

$$\omega_p = e^{i\omega_0}.$$

Further, for each integer k , we define the discrete-time complex exponential

$$\psi_k[n] = \omega_p^{kn} = e^{i\omega_0 kn}, \quad n \in \mathbb{Z}.$$

Each ψ_k is p -periodic (prove this!) so we can think of it as a vector in $\mathbb{C}^p$ containing only one period:

$$\psi_k = [\psi_k[0] \quad \psi_k[1] \quad \cdots \quad \psi_k[p-1]]^\top.$$

Convince yourself that this is sufficient to describe ψ_k .

(a) For which integers $k \in \mathbb{Z}$ is $\omega_p^k = 1$? Use your answer to show that

$$\psi_{k+p}[n] = \psi_k[n] \quad \text{for all } n \in \mathbb{Z},$$

so that among $\{\psi_k : k \in \mathbb{Z}\}$ there are only p distinct signals, namely $\psi_0, \psi_1, \dots, \psi_{p-1}$.

Solution: $\omega_p^k = 1$ when k is a multiple of p . This is because

$$\omega_p^k = e^{i\omega_0 k} = \cos(\omega_0 k) + i \sin(\omega_0 k),$$

so we have $\omega_p^k = 1$ exactly when $\frac{2\pi k}{p}$ is a multiple of 2π , i.e., when $k = mp$ for an integer m .

Now consider that we can write any integer as $k = mp + r$ where r , the remainder after dividing k by p , is necessarily one of $0, 1, \dots, p-1$. Hence

$$\psi_k[n] = \psi_{mp+r}[n] = \omega_p^{(mp+r)n} = \omega_p^{mpn} \omega_p^{rn} = \psi_r[n] \cdot 1 = \psi_r[n],$$

so any ψ_k is equivalent to ψ_r for some $r \in 0, 1, \dots, p-1$. Finally, recall that two signals are *equal* only when they agree for *all* n . Among the ψ_r , it can be shown that none are equivalent by considering their value at, say, $n = 1$. This shows that there are exactly p distinct signals among $\{\psi_k : k \in \mathbb{Z}\}$.

(b) Let $m \in \mathbb{Z}$. Show that

$$\sum_{n=0}^{p-1} \omega_p^{mn} = \begin{cases} p & \text{if } m \text{ is a multiple of } p, \\ 0 & \text{otherwise.} \end{cases}$$

Hint: use the geometric-series sum formula.

Solution: **Case 1: m is a multiple of p .** By part (a), $\omega_p^m = 1$, so every term of the sum is $(\omega_p^m)^n = 1$ and the sum is p .

Case 2: m is not a multiple of p . Let $r = \omega_p^m$. By part (a), $r \neq 1$, so the geometric-series formula applies (it requires $r \neq 1$, otherwise we would be dividing by zero):

$$\sum_{n=0}^{p-1} r^n = \frac{1-r^p}{1-r} = \frac{1-\omega_p^{mp}}{1-\omega_p^m} = \frac{1-(\omega_p^p)^m}{1-\omega_p^m} = \frac{1-1}{1-\omega_p^m} = 0.$$

The special case $m = 1$ says the p -th roots of unity $\omega_p^0, \omega_p^1, \dots, \omega_p^{p-1}$ sum to zero: they are equally spaced around the unit circle, so they cancel.

- (c) Show that $\langle \psi_k, \psi_\ell \rangle = 0$ whenever $k \neq \ell$ with $k, \ell \in \{0, 1, \dots, p-1\}$, and compute $\|\psi_k\|^2 = \langle \psi_k, \psi_k \rangle$. We repeat the inner product of complex-valued vectors here for your convenience:

$$\langle u, v \rangle = \sum_{n=0}^{p-1} u[n] v[n]^*.$$

Solution: Since $|\omega_p| = 1$, we have $(\omega_p^{\ell n})^* = \omega_p^{-\ell n}$, so

$$\langle \psi_k, \psi_\ell \rangle = \sum_{n=0}^{p-1} \omega_p^{kn} \omega_p^{-\ell n} = \sum_{n=0}^{p-1} \omega_p^{(k-\ell)n}.$$

This is the sum from part (b) with $m = k - \ell$. If $k \neq \ell$, then $0 < |k - \ell| \leq p - 1$, so $k - \ell$ is not a multiple of p and the sum is 0. If $k = \ell$, then $m = 0$ is a multiple of p and the sum is p . Hence

$$\langle \psi_k, \psi_\ell \rangle = \begin{cases} p & k = \ell, \\ 0 & k \neq \ell, \end{cases} \quad \text{and in particular} \quad \|\psi_k\|^2 = p.$$

The signals $\psi_0, \dots, \psi_{p-1}$ are mutually orthogonal, each with norm $\sqrt{p}$.

- (d) Prove that $\{\psi_0, \psi_1, \dots, \psi_{p-1}\}$ is linearly independent, and conclude that it is a basis for $\mathbb{C}^p$.

Hint: try taking the inner product of $\sum_{k=0}^{p-1} a_k \psi_k = \mathbf{0}$ with an arbitrary ψ_ℓ .

Solution: Suppose $a_0, \dots, a_{p-1} \in \mathbb{C}$ satisfy

$$\sum_{k=0}^{p-1} a_k \psi_k = \mathbf{0}.$$

Fix any $\ell \in \{0, \dots, p-1\}$ and take the inner product of both sides with ψ_ℓ . Using linearity of the inner product in its first argument and part (c),

$$0 = \left\langle \sum_{k=0}^{p-1} a_k \psi_k, \psi_\ell \right\rangle = \sum_{k=0}^{p-1} a_k \langle \psi_k, \psi_\ell \rangle = a_\ell \|\psi_\ell\|^2 = p a_\ell.$$

Since $p \neq 0$, $a_\ell = 0$. This holds for every ℓ , so the only linear combination giving $\mathbf{0}$ is the trivial one: the set is linearly independent. (The same argument shows that *any* collection of nonzero, mutually orthogonal vectors is linearly independent.)

We now have p linearly independent vectors in the p -dimensional space $\mathbb{C}^p$, so they span $\mathbb{C}^p$ and form a basis. This is the *discrete Fourier basis*.

- (e) Consider the real-valued signal

$$x[n] = 4 \cos\left(\frac{\pi}{4}n\right) + 2 \sin\left(\frac{\pi}{2}n\right) + 5.$$

For this part only, let $p = 8$. Write x as a linear combination of the signals ψ_k :

$$x[n] = \sum_{k=0}^7 X_k \psi_k[n].$$

In particular, you should clearly identify all eight coefficients $X_0, \dots, X_7$. Then explain, using part (d), why this choice of coefficients is *unique*.

Hint: using part (a), you can write $\psi_{-k} = \psi_{mp-k}$ for any integer m . For example, with $p = 8$, we can choose $m = 1$ to rewrite $\psi_{-1} = \psi_7$.

Solution: By the inverse Euler formulas $\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$, and using $\frac{1}{i} = -i$,

$$x[n] = 2e^{i\omega_0 n} + 2e^{-i\omega_0 n} - i e^{i2\omega_0 n} + i e^{-i2\omega_0 n} + 5.$$

With $p = 8$ we have $\omega_0 = \frac{2\pi}{8} = \frac{\pi}{4}$ and $\psi_k[n] = e^{i\omega_0 kn}$, so

$$e^{i\omega_0 n} = \psi_1[n], \quad e^{i2\omega_0 n} = \psi_2[n], \quad e^{-i\omega_0 n} = \psi_{-1}[n] = \psi_7[n], \quad e^{-i2\omega_0 n} = \psi_{-2}[n] = \psi_6[n],$$

where the last two use $\psi_{k+8} = \psi_k$ from part (a). The constant term is also a basis signal: $5 = 5 \cdot e^{i \cdot 0 \cdot n} = 5 \psi_0[n]$. Therefore

$$x[n] = 5\psi_0[n] + 2\psi_1[n] - i\psi_2[n] + i\psi_6[n] + 2\psi_7[n],$$

i.e.,

$$X_0 = 5, \quad X_1 = 2, \quad X_2 = -i, \quad X_6 = i, \quad X_7 = 2, \quad X_3 = X_4 = X_5 = 0.$$

Notice that $X_7 = X_{-1} = X_1^*$ and $X_6 = X_{-2} = X_2^*$: the coefficients of a real-valued signal come in conjugate pairs at frequencies $\pm k$. The constant (“DC”) part of the signal lives entirely in X_0 , since $\psi_0[n] = 1$ for all n .

Uniqueness. Suppose both $\{X_k\}$ and $\{\tilde{X}_k\}$ represent x . Subtracting the two representations,

$$\sum_{k=0}^7 (X_k - \tilde{X}_k) \psi_k = x - x = \mathbf{0}.$$

By the linear independence proved in part (d), every coefficient of this combination must be zero, so $X_k = \tilde{X}_k$ for all k . The decomposition above is the *only* one. This is what makes the phrase “the frequency content of x ” meaningful: because the representation is unique, the amount of x at frequency $\frac{2\pi k}{p}$ is a well-defined number X_k , not an artifact of how we happened to write x down.

3. Matching Frequencies

(a) Let $\omega \in \mathbb{R}$ be a fixed frequency and $\phi_1, \phi_2 \in \mathbb{R}$ be phases. Using Euler's identity, show that

$$\cos(\omega t + \phi_1) + \cos(\omega t + \phi_2)$$

can be written as a single cosine of the same frequency in the form

$$A \cos(\omega t + \phi).$$

and derive expressions for A and ϕ .

Hint: You may find the identity $e^{i\phi_1} + e^{i\phi_2} = e^{i\left(\frac{\phi_1+\phi_2}{2}\right)} \left(e^{i\left(\frac{\phi_1-\phi_2}{2}\right)} + e^{-i\left(\frac{\phi_1-\phi_2}{2}\right)} \right)$ useful.

Solution: Using Euler's identity, we know we can rewrite

$$\cos(\theta) = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$$

$$\begin{aligned} \cos(\omega t + \phi_1) + \cos(\omega t + \phi_2) &= \frac{1}{2} \left(e^{i(\omega t + \phi_1)} + e^{-i(\omega t + \phi_1)} + e^{i(\omega t + \phi_2)} + e^{-i(\omega t + \phi_2)} \right) \\ &= \frac{1}{2} \left((e^{i\phi_1} + e^{i\phi_2}) e^{i\omega t} + (e^{-i\phi_1} + e^{-i\phi_2}) e^{-i\omega t} \right) \end{aligned}$$

We can simplify $e^{i\phi_1} + e^{i\phi_2}$ by factoring out the average, which is a common pattern.

$$\begin{aligned} e^{i\phi_1} + e^{i\phi_2} &= e^{i\left(\frac{\phi_1+\phi_2}{2}\right)} \left(e^{i\left(\frac{\phi_1-\phi_2}{2}\right)} + e^{-i\left(\frac{\phi_1-\phi_2}{2}\right)} \right) \\ &= e^{i\left(\frac{\phi_1+\phi_2}{2}\right)} \left(2 \cos\left(\frac{\phi_1-\phi_2}{2}\right) \right) \end{aligned}$$

We can apply the same trick to simplify

$$e^{-i\phi_1} + e^{-i\phi_2} = 2 \cos\left(\frac{\phi_1-\phi_2}{2}\right) e^{-i\left(\frac{\phi_1+\phi_2}{2}\right)}$$

Continuing to simplify using these new expressions,

$$\begin{aligned} \cos(\omega t + \phi_1) + \cos(\omega t + \phi_2) &= \cos\left(\frac{\phi_1-\phi_2}{2}\right) \left(e^{i\omega t\left(\frac{\phi_1+\phi_2}{2}\right)} + e^{-i\omega t\left(\frac{\phi_1+\phi_2}{2}\right)} \right) \\ &= 2 \cos\left(\frac{\phi_1-\phi_2}{2}\right) \cos\left(\omega t + \left(\frac{\phi_1+\phi_2}{2}\right)\right) \end{aligned}$$

Thus, we've demonstrated that the sum of two out-of-phase cosines with the same frequency can be written as a cosine of a single phase at that frequency, where

$$A = 2 \cos\left(\frac{\phi_1-\phi_2}{2}\right), \quad \phi = \frac{\phi_1+\phi_2}{2}$$

- (b) In fact, any linear combination of a set of sinewaves of frequency ω is always a sinewave of the same frequency, *even if* each has a distinct phase and amplitude. Prove this. In particular, show that

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) = A \cos(\omega t + \phi).$$

Hint: it might be helpful to recognize that

$$A \cos(\omega t + \phi) = \sum_{k=1}^N A_k \cos(\omega t + \phi_k)$$

can be obtained by equating the real parts of the two sides of

$$A e^{i(\omega t + \phi)} = \sum_{k=1}^N A_k e^{i(\omega t + \phi_k)}.$$

Note that it is insufficient to assert that the above statement is true; rather, this is equivalent to and perhaps easier-to-prove than your goal. In other words, proving that the complex exponential equality above holds proves that the cosine equality holds.

Solution: Following the hint, we can work with *complex exponentials*, and eventually take the real part to prove the desired cosine identity:

$$\sum_{k=1}^N A_k e^{i(\omega t + \phi_k)} = A e^{i(\omega t + \phi)}.$$

Since we can't assume this equality to be true (this is what we want to *prove* to be true), we start with the left-hand side and factor out $e^{i\omega t}$ (which is never zero), yielding

$$\sum_{k=1}^N A_k e^{i(\omega t + \phi_k)} = e^{i\omega t} \sum_{k=1}^N A_k e^{i\phi_k}.$$

The remaining sum $\sum_{k=1}^N A_k e^{i\phi_k}$ is *simply a complex number*, so it can be written in polar form as $A e^{i\phi}$ for *some* suitable real A and ϕ . You don't need to know A and ϕ ; what's important is that they *exist*! Substituting this polar form back in,

$$\sum_{k=1}^N A_k e^{i(\omega t + \phi_k)} = A e^{i(\omega t + \phi)}.$$

Using Euler's formula $e^{i\theta} = \cos(\theta) + i \sin(\theta)$ and expanding both sides yields

$$\sum_{k=1}^N A_k (\cos(\omega t + \phi_k) + i \sin(\omega t + \phi_k)) = A (\cos(\omega t + \phi) + i \sin(\omega t + \phi)).$$

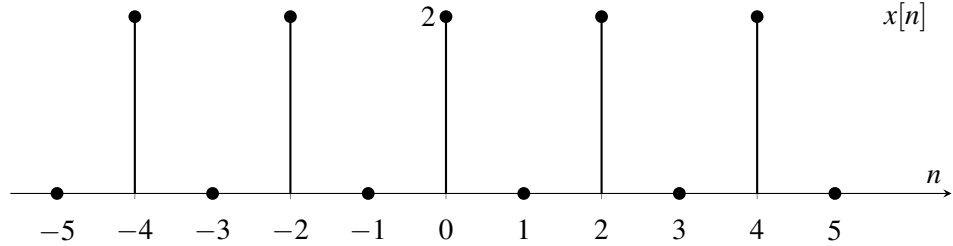
Equating real parts gives

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) = A \cos(\omega t + \phi),$$

which is the desired identity.

4. DTFS Warmup (Adapted from EE 120 Discussion)

Consider the periodic DT signal $x[n]$ shown below:



Throughout this problem, we explore the complex exponential discrete-time Fourier series (DTFS) for $x[n]$.

- (a) Determine the fundamental period p and fundamental frequency ω_0 of $x[n]$.

Solution:

The fundamental period of a periodic DT signal $x[n]$ is the minimum positive integer p such that

$$x[n+p] = x[n], \quad \forall n \in \mathbb{Z} \quad (1)$$

By inspection, we see $x[n]$ repeats every 2 time steps, so $p = 2$. To find the fundamental frequency, we use the formula $\omega = \frac{2\pi}{p}$ (radians per unit time).

$$\omega_0 = \frac{2\pi}{2} = \pi \quad (2)$$

- (b) Determine an expression of the signal $x[n]$ as a linear combination of complex exponentials. Do this by inspection, without resorting to the machinery of the DTFS analysis or synthesis equations.

Solution:

We can observe that $x[n]$ can be expressed as

$$x[n] = (-1)^n + 1 = e^{i\pi n} + e^0$$

- (c) Now use the analysis equation to determine the DTFS representation of $x[n]$. That is, determine X_k in the DTFS expansion. For your convenience, the analysis equation is repeated below:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}. \quad (3)$$

You should use the p and ω_0 found in part (a).

Solution:

Because our period is $p = 2$, we can choose the period $\{0, 1\}$ to be our period of choice for analysis. Therefore, our DTFS coefficients are:

$$\begin{aligned} X_0 &= \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-i(0)\omega_0 n} = \frac{1}{2} \left(x[0] e^{-i(0)\pi(0)} + x[1] e^{-i(0)\pi(1)} \right) = \frac{1}{2} (2e^0 + 0e^0) = 1 \\ X_1 &= \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-i(1)\omega_0 n} = \frac{1}{2} \left(x[0] e^{-i(1)\pi(0)} + x[1] e^{-i(1)\pi(1)} \right) = \frac{1}{2} (2e^0 + 0e^{-i\pi}) = 1 \end{aligned}$$

- (d) As a sanity check, use the DTFS coefficients X_k to determine the equation for $x[n]$, and confirm that it matches your answer for part (b). You may find the synthesis equation helpful:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad (4)$$

Hint: Try splitting up the problem by solving for $x[n]$ when n is even and when it is odd.

Solution:

Let $\ell \in \mathbb{Z}$ be an integer value. It is the case that 2ℓ is always even and $2\ell + 1$ is always odd. Using this, we can see that for all ℓ :

$$\begin{aligned} x[2\ell] &= \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} = \sum_{k=0}^1 (1) e^{ik\pi(2\ell)} = \sum_{k=0}^1 e^{i2\pi k\ell} = \sum_{k=0}^1 1 = 1 + 1 = 2 \\ x[2\ell + 1] &= \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} = \sum_{k=0}^1 (1) e^{ik\pi(2\ell+1)} = \sum_{k=0}^1 e^{i2\pi k\ell + ik\pi} = \sum_{k=0}^1 e^{i2\pi k\ell} e^{ik\pi} = (1)(1) + (1)(-1) = 0 \end{aligned}$$

We have demonstrated that under our assignments of the DTFS coefficients, $x[n] = 2$ for all even n and $x[n] = 0$ for all odd n , which is precisely the pattern we see in the signal.

Alternatively, we could solve specifically for the values of $x[0]$ and $x[1]$, and then argue that their values continue periodically across the whole signal.

5. Signal Analysis with Orthogonal Projections

In many applications, signals are mixtures of a few frequencies. In this problem, you will project a length- p signal onto complex-exponential basis vectors and analyze the resultant coefficients. We define $\omega_0 = \frac{2\pi}{p}$ for the fundamental frequency.

This problem has an IPython component. Please attach a PDF printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

(a) Write two Python functions:

- (i) `complex_inner_product(u, v)`: takes two complex vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^p$ and returns $\langle \mathbf{u}, \mathbf{v} \rangle$.
- (ii) `generate_basis_vector(k, p)`: returns the DTFS basis vector $\boldsymbol{\psi}_k$ of length p with entries $\boldsymbol{\psi}_k[n] = e^{i\omega_0 nk}$.
- (iii) Using your functions, with $p = 64$, numerically evaluate $\langle \boldsymbol{\psi}_3, \boldsymbol{\psi}_{10} \rangle$ and report its magnitude.

Solution: Your functions should implement the definitions verbatim: $\sum_{n=0}^{p-1} u[n]v[n]^*$ for the inner product, and compute $e^{i\omega_0 nk}$ for each component of $\boldsymbol{\psi}_k$. Orthogonality implies $\langle \boldsymbol{\psi}_3, \boldsymbol{\psi}_{10} \rangle = 0$ in exact arithmetic, but numerically, the magnitude might be slightly off 0 due to floating point arithmetic. See the solution IPython notebook for an example of a correct implementation.

(b) Let $p = 64$, $f_1 = 2$ Hz, $f_2 = 8$ Hz, $A_1 = 2.0$, $A_2 = 0.5$. Define $t_n = n/p$ and $s: \mathbb{R} \rightarrow \mathbb{R}$ as

$$s(t) = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t).$$

Write a Python function `generate_signal(p, frequencies, amplitudes)` that returns $\mathbf{s} \in \mathbb{R}^{64}$ with entries $s[n] = s(t_n)$ for these parameters. Additionally report the first four samples $s[0:4]$. Note that instead of s being a discrete-time signal, it is a continuous-time signal for which we take discrete samples, which gives us the vector $\mathbf{s}$.

Solution: Compute $t_n = n/64$ and evaluate the cosine sum at each t_n to obtain a length-64 real vector. See the solution IPython notebook for an example of a correct implementation.

(c) The contribution of frequency index k in $\mathbf{s}$ is given by the projection coefficient

$$c_k = \text{proj}_{\boldsymbol{\psi}_k}(\mathbf{s}) = \frac{\langle \mathbf{s}, \boldsymbol{\psi}_k \rangle}{\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_k \rangle} = \frac{1}{p} \langle \mathbf{s}, \boldsymbol{\psi}_k \rangle.$$

Write a Python function `get_projection_coefficient(signal, k, p)` that computes c_k using your tools from part (a).

Solution: Implement $c_k = \frac{1}{p} \sum_{n=0}^{p-1} s[n] \boldsymbol{\psi}_k[n]^*$. See the solution IPython notebook for an example of a correct implementation.

(d) For $p = 64$, $k = 0, 1, \dots, 31$, compute c_k and plot $|c_k|$ versus k , with labeled axes. Are there peaks at $k = 2$ and $k = 8$? How do the magnitudes relate to A_1 and A_2 ?

Solution: With $p = 64$ and $f_1 = 2$, $f_2 = 8$, peaks should occur at $k = 2$ and $k = 8$. Magnitudes reflect A_1 and A_2 but halved. See the solution IPython notebook for an example of a correct implementation and plot.

(e) Repeat parts (b) through (d) with the following frequencies and amplitudes: $f_1 = 24$, $f_2 = 26$, $A_1 = 1$, $A_2 = 1$. We will refer to this signal as s_2 . Afterwards, generate a plot of the discrete-time signal $s_2[n]$. Describe the signal in words. Does the signal match what you'd expect from the DTFS coefficients?

Solution: With $p = 64$ and $f_1 = 24$, $f_2 = 26$, peaks should occur at $k = 24$ and $k = 26$. Magnitudes

reflect A_1 and A_2 but halved, as before. When plotting the discrete-time signal, we see that while it is a sum of two cosines of different frequencies, the resultant signal is a 25 Hz signal modulated by a 2 Hz signal. Neither of these frequencies is present in the DTFS coefficients. See the solution IPython notebook for an example of a correct implementation and plot.

6. DTFS Round Trip

Recall the DTFS analysis and synthesis equations:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n}, X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}.$$

For every subpart, let $p = 4$.

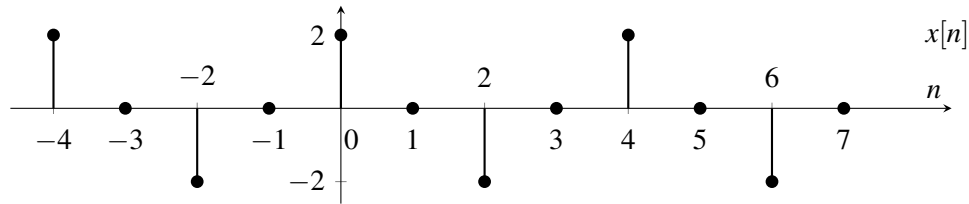
- (a) Determine ω_0 and write out $\psi_0, \psi_1, \psi_2, \psi_3$ explicitly as vectors in $\mathbb{C}^4$.

Solution: $\omega_0 = 2\pi/p = \pi/2$, so $\psi_k[n] = i^{kn}$:

$$\psi_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \psi_1 = \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}, \quad \psi_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \psi_3 = \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix}.$$

We have seen and worked with this exact set of vectors extensively, long before learning of DTFS!

- (b) Consider the 4-periodic signal $x[n]$ shown below. Compute its DTFS coefficients X_0, X_1, X_2, X_3 .



Solution: Using the analysis equation with $e^{-ik\omega_0 n} = (-i)^{kn}$, i.e., $X_k = \frac{1}{4} \langle x, \psi_k \rangle$:

$$X_0 = \frac{1}{4} (2 + 0 - 2 + 0) = 0$$

$$X_1 = \frac{1}{4} (2 \cdot 1 + 0 \cdot (-i) + (-2) \cdot (-1) + 0 \cdot i) = \frac{1}{4} (2 + 2) = 1$$

$$X_2 = \frac{1}{4} (2 \cdot 1 + 0 \cdot (-1) + (-2) \cdot 1 + 0 \cdot (-1)) = \frac{1}{4} (2 - 2) = 0$$

$$X_3 = \frac{1}{4} (2 \cdot 1 + 0 \cdot i + (-2) \cdot (-1) + 0 \cdot (-i)) = \frac{1}{4} (2 + 2) = 1$$

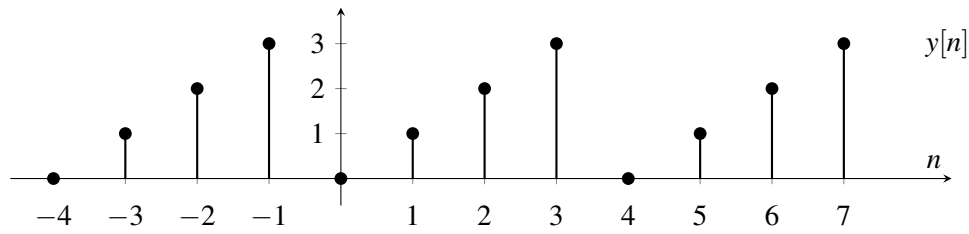
- (c) Use the synthesis equation first to write $x[n]$ as a sum of complex exponentials. Then simplify your answer to write $x[n]$ in *real* form, i.e., as a constant plus real-valued sinusoids.

Solution:

$$\begin{aligned} x[n] &= X_1 \psi_1[n] + X_3 \psi_3[n] \\ &= e^{i\frac{\pi}{2}n} + e^{-i\frac{\pi}{2}n} \\ &= 2 \cos\left(\frac{\pi}{2}n\right). \end{aligned}$$

Check:

$$\begin{aligned} x[0] &= 2 \cos(0) = 2 \cdot 1 = 2, \\ x[1] &= 2 \cos\left(\frac{\pi}{2}\right) = 2 \cdot 0 = 0, \\ x[2] &= 2 \cos(\pi) = 2 \cdot (-1) = -2, \\ x[3] &= 2 \cos\left(\frac{3\pi}{2}\right) = 2 \cdot 0 = 0. \end{aligned}$$



(d) Now let $y[n]$ be the 4-periodic “ramp” shown below. Compute its DTFS coefficients Y_0, Y_1, Y_2, Y_3 .

Solution: Using the analysis equation as in part (b):

$$\begin{aligned} Y_0 &= \frac{1}{4} (0 + 1 + 2 + 3) = \frac{3}{2} \\ Y_1 &= \frac{1}{4} (0 \cdot 1 + 1 \cdot (-i) + 2 \cdot (-1) + 3 \cdot i) = \frac{1}{4} (-2 + 2i) = \frac{-1+i}{2} \\ Y_2 &= \frac{1}{4} (0 - 1 + 2 - 3) = -\frac{1}{2} \\ Y_3 &= \frac{1}{4} (0 \cdot 1 + 1 \cdot i + 2 \cdot (-1) + 3 \cdot (-i)) = \frac{1}{4} (-2 - 2i) = \frac{-1-i}{2} \end{aligned}$$

(e) As in part (c), write $y[n]$ as a sum of complex exponentials and then in real form.

Hint: You should have noticed that $Y_3 = Y_1^$ (if not, go back and check your work!). Try to express Y_1 as a complex exponential, i.e., write $Y_1 = |Y_1| e^{i\angle Y_1}$ and $Y_3 = Y_1^* = |Y_1| e^{-i\angle Y_1}$. These facts will help you combine the $k = 1$ and $k = 3$ terms to create a single cosine.*

Solution: The synthesis equation gives

$$\begin{aligned} y[n] &= Y_0 \psi_0[n] + Y_1 \psi_1[n] + Y_2 \psi_2[n] + Y_3 \psi_3[n] \\ &= \frac{3}{2} + \frac{-1+i}{2} e^{i\frac{\pi}{2}n} - \frac{1}{2} e^{i\pi n} + \frac{-1-i}{2} e^{-i\frac{\pi}{2}n}. \end{aligned}$$

Following the hint, we write Y_1 in polar form. The magnitude is

$$|Y_1| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}.$$

You could determine $\angle Y_1$ by drawing Y_1 in the complex plane. Alternatively, you could compute $\angle Y_1$ by noticing that the real part $\text{Re}(Y_1) = -\frac{1}{2}$ is negative and the imaginary part $\text{Im}(Y_1) = \frac{1}{2}$ is positive, so Y_1 lies in the second quadrant. Since the reference angle is

$$\tan^{-1}\left(\frac{|\text{Im}(Y_1)|}{|\text{Re}(Y_1)|}\right) = \tan^{-1}(1) = \frac{\pi}{4},$$

the true angle is

$$\angle Y_1 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}.$$

As a quick confirmation, $e^{i\frac{3\pi}{4}} = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$, so

$$|Y_1| e^{i\angle Y_1} = \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\frac{1}{2} + i\frac{1}{2} = Y_1.$$

Thus

$$\begin{aligned} Y_1 &= \frac{\sqrt{2}}{2} e^{i\frac{3\pi}{4}}, \\ Y_3 &= Y_1^* = \frac{\sqrt{2}}{2} e^{-i\frac{3\pi}{4}}. \end{aligned}$$

Since $\psi_3[n] = e^{-i\frac{\pi}{2}n}$, the $k = 1$ and $k = 3$ terms combine into a single cosine:

$$\begin{aligned} Y_1 \psi_1[n] + Y_3 \psi_3[n] &= \frac{\sqrt{2}}{2} e^{i(\frac{\pi}{2}n + \frac{3\pi}{4})} + \frac{\sqrt{2}}{2} e^{-i(\frac{\pi}{2}n + \frac{3\pi}{4})} \\ &= \sqrt{2} \cos\left(\frac{\pi}{2}n + \frac{3\pi}{4}\right). \end{aligned}$$

Adding the $k = 0$ term $Y_0 \psi_0[n] = \frac{3}{2}$ and the $k = 2$ term $Y_2 \psi_2[n] = -\frac{1}{2}(-1)^n$,

$$y[n] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{\pi}{2}n + \frac{3\pi}{4}\right) - \frac{1}{2}(-1)^n.$$

Check:

$$y[0] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{3\pi}{4}\right) - \frac{1}{2} = \frac{3}{2} + \sqrt{2} \left(-\frac{\sqrt{2}}{2}\right) - \frac{1}{2} = \frac{3}{2} - 1 - \frac{1}{2} = 0,$$

$$y[1] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{5\pi}{4}\right) + \frac{1}{2} = \frac{3}{2} + \sqrt{2} \left(-\frac{\sqrt{2}}{2}\right) + \frac{1}{2} = \frac{3}{2} - 1 + \frac{1}{2} = 1,$$

$$y[2] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{7\pi}{4}\right) - \frac{1}{2} = \frac{3}{2} + \sqrt{2} \left(\frac{\sqrt{2}}{2}\right) - \frac{1}{2} = \frac{3}{2} + 1 - \frac{1}{2} = 2,$$

$$y[3] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{9\pi}{4}\right) + \frac{1}{2} = \frac{3}{2} + \sqrt{2} \left(\frac{\sqrt{2}}{2}\right) + \frac{1}{2} = \frac{3}{2} + 1 + \frac{1}{2} = 3.$$

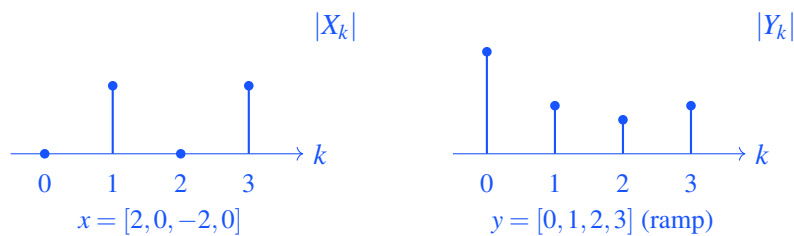
Unlike part (c), some of the coefficients here are complex, and the phase $\angle Y_1 = \frac{3\pi}{4}$ shows up as a phase shift inside the cosine.

- (f) For each of the two signals above, sketch a stem plot of $|X_k|$ (respectively, $|Y_k|$) versus k . Notice that some of the X_k s are 0, whereas all of the Y_k s are non-zero (if not, go back and check your work!). Using the original signals, justify this observation.

In particular, you should compare characteristics of the original signal which explain the fact that $y[n]$'s DTFS expansion has a constant term whereas $x[n]$'s does not. Make a similar argument to explain why $y[n]$'s DTFS expansion needs the fastest-varying basis signal $(-1)^n$ but $x[n]$ does not.

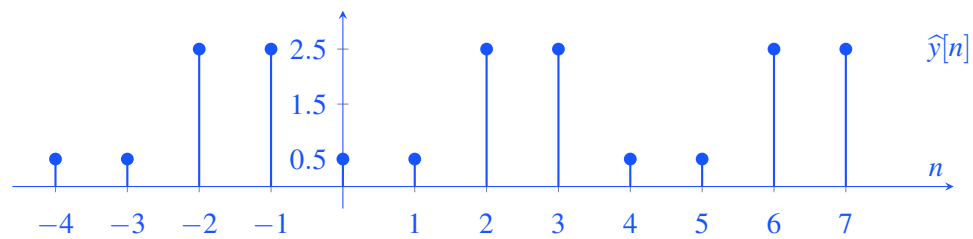
Hint: try plotting the signal you'd get by removing the ψ_2 term from $y[n]$'s DTFS expansion. What does this look like? In general, what contribution does a rapidly oscillating term like $(-1)^n$ make to a signal's appearance?

Solution: We are plotting the *magnitude* of the coefficients. Respectively, the coefficient magnitudes are $|X_k| = [0, 1, 0, 1]$ and $|Y_k| = [\frac{3}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}]$. Plotted, these look like:



The constant term in a signal's DTFS expansion represents its average value. Clearly, $x[n]$ has an average of 0 whereas the ramp's average value is positive. This is why $X_0 = 0$ but $Y_0 \neq 0$.

The ramp y climbs 0, 1, 2, 3 and then jumps back down to 0 at $n = 4$ before quickly jumping back up to 1. These sharp changes require the fastest basis signal $\psi_2[n] = (-1)^n$. We can try plotting the result *without* this term, $\hat{y}[n] = \frac{3}{2} + \sqrt{2} \cos\left(\frac{\pi}{2}n + \frac{3\pi}{4}\right)$:



We see the overall *shape* of the ramp, but some "fine detail" has been lost. More generally, removing the high frequency components of *any* signal will result in a "smoothed" appearance. Looking at $x[n]$, there is less rapid oscillation, so it makes sense that there is no $(-1)^n$ component. More specifically, $x[n]$ are samples of a single cosine, so it already lives entirely in the $k = \pm 1$ terms and has no need for $k = 0$ or $k = 2$.

7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) We just finished Exam 1. Congrats! We would love to hear some feedback while the experience is “fresh.” Perhaps you already had a chance to share your thoughts with course staff during discussion or office hours. Regardless (in case you did not, or if they forgot what you shared) we are hoping you will write something down here. We will do what we can to make the next exam better!

Solution:

Most of the problems felt similar to lecture and discussion, but there was a problem about _____ that I felt totally unprepared for. I am wondering if there is something I could have done to prepare better for this type of problem.

The exam started really late because there was not enough staff to hand out exams, and it took a really long time for them to count the exams at the end. We ended up staying in the room for 30 minutes past the expected end time!

- (b) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

- (c) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

- (d) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

- (e) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able to practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

- (f) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.